



DEPARTMENT OF MATHEMATICS
NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

B.Tech. Second Semester (CSE/IT/AIML/IOT/MC/MV/ADS/PI/SET) Mid Term – II Examination

Date: April 07, 2025

Max. Marks: 15

Time: 10:00 am – 10:50 am

Paper: Integral Calculus and Difference Equations (MAIC 102)

- Instructions:**
- Answer all the questions.
 - The calculator is not allowed into the examination hall.
 - Start writing your answers from the first page. Do not leave it blank. No extra answer sheets will be provided.

1. By transforming into cylindrical co-ordinates evaluate the integral $\iiint (x^2 + y^2 + z^2) dx dy dz$ taken over the region $0 \leq z \leq x^2 + y^2 \leq 1$. [5M]
2. Find the constants a and b so that the surface $ax^2 - byz = (a + 2)x$ will be orthogonal to the surface $4x^2y + z^3 = 4$ at the point $(1, -1, 2)$. [5M]
3. Find the work done by the force $\vec{F} = (x^2 - y^3)\hat{i} + (x + y)\hat{j}$ in moving a particle along the closed path C containing the curves $x + y = 0$, $x^2 + y^2 = 16$ and $y = x$ in the first and fourth quadrants. [5M]